

siRNA Mediated Knockdown of FUT7 to Attenuate Sialyl Lewis X Driven Triple-Negative Breast Cancer Metastasis

Introduction

Triple-Negative Breast Cancer

Triple-negative breast cancer (TNBC) is **highly metastatic** (Tierno et al., 2023).

Hormone based therapies are **ineffective** due to lack of target receptors (Yin et al., 2020).

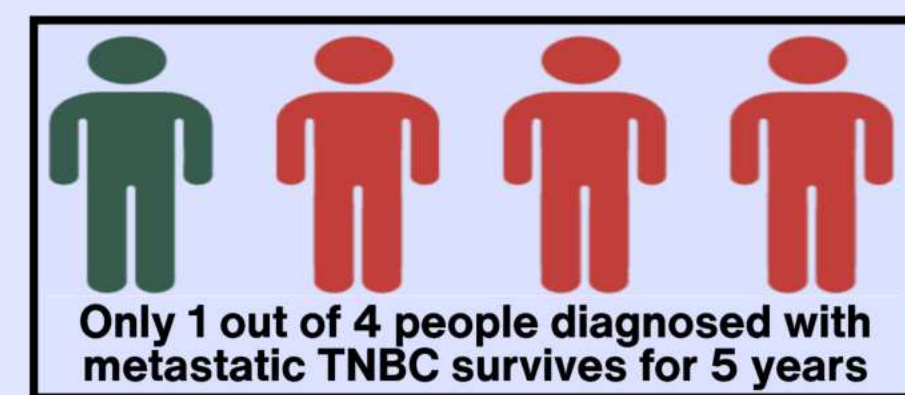


Figure 1. Five-year survival rate of metastatic TNBC, illustrating its poor prognosis. Graphic created by researcher using BioRender, 2026

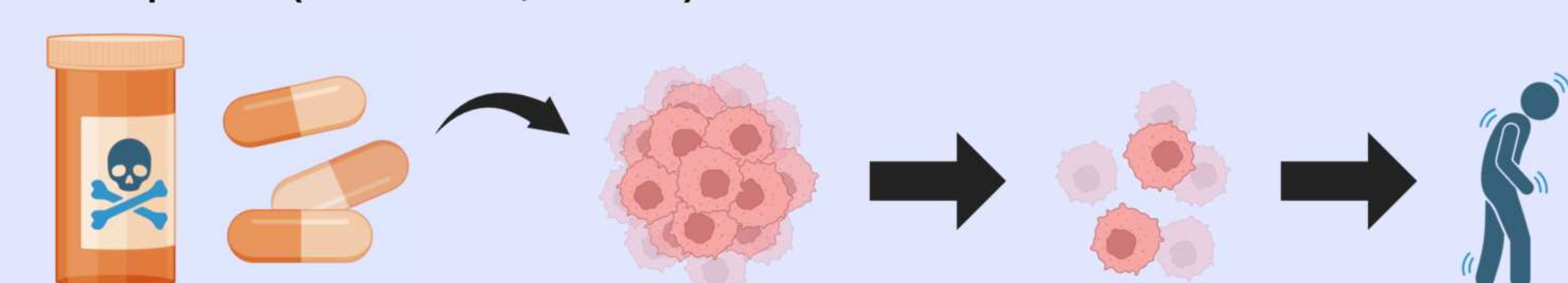
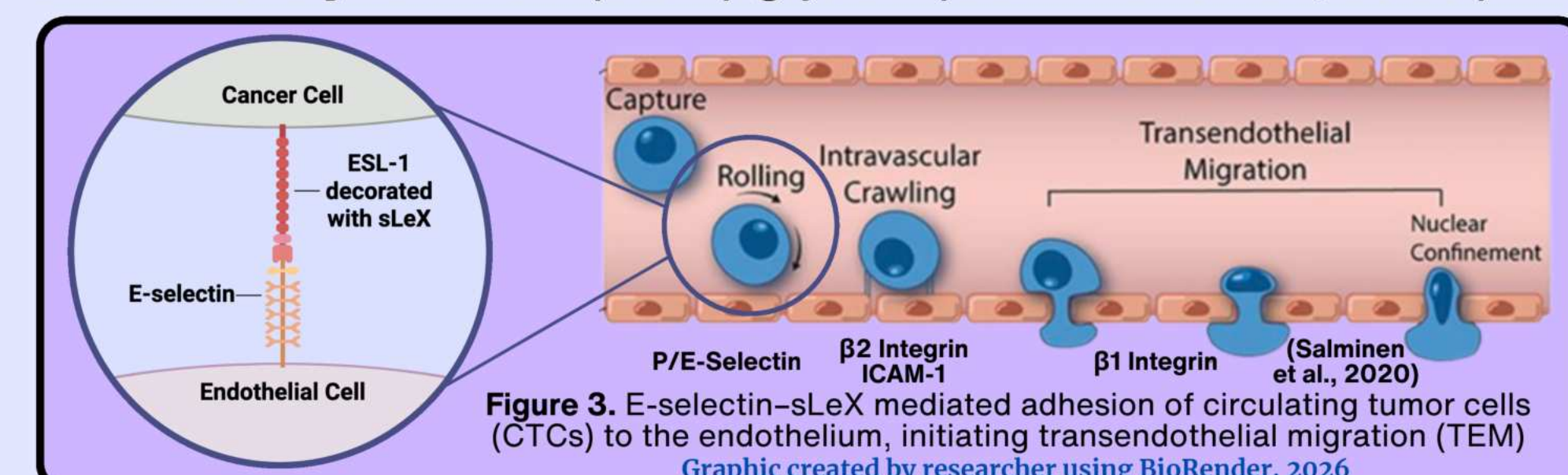


Figure 2. Schematic showing how circulating tumor cells (CTCs) escape conventional therapies, circulate through the bloodstream, and form metastatic tumors. Graphic created by researcher using BioRender, 2026

Current therapies eliminate proliferating tumor cells but may leave **circulating tumor cells (CTCs)**, which can travel through the bloodstream and **seed metastases** (Wang et al., 2025).

Transendothelial Migration

CTCs **exit the bloodstream** via the transendothelial migration cascade (TEM), mediated by adhesion between **E-selectin** and the **sialyl Lewis X (sLeX)** glycan (Salminen et al., 2020).



Without this initial interaction, cancerous cells would not be able to slow down enough to continue the remainder of the TEM and would be trapped within blood vessels to **eventually die due to blood shear**, preventing metastasis (Connolly et al., 2021).

Project Overview

Research Question

Can reducing sLeX expression limit breast cancer metastasis and serve as a potential treatment for cancer?

Hypothesis

If 4T1 cells are transfected with FUT7 siRNA, then the invasive nature of the cells will significantly decrease.

Justification

Aberrant **glycosylation of sLeX** is a common feature in cancer and significantly **contributes to TEM-mediated metastasis** (Aghamiri & Amin, 2024). Additionally, it is known that FUT7 plays a large role in the creation of sLeX (Qin et al., 2020).

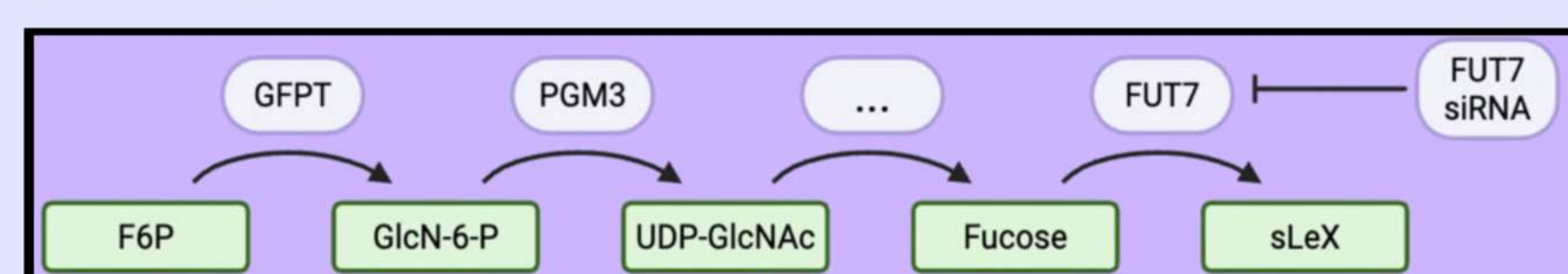


Figure 4. Pathway of the creation of sLeX. Graphic created by researcher using BioRender, 2026

The functional contribution of sLeX to local invasion and E-selectin-mediated adhesion has not been fully evaluated; this study addresses that gap and introduces a PROTAC-based strategy targeting FUT7 to reduce glycan driven metastasis.



Methodology

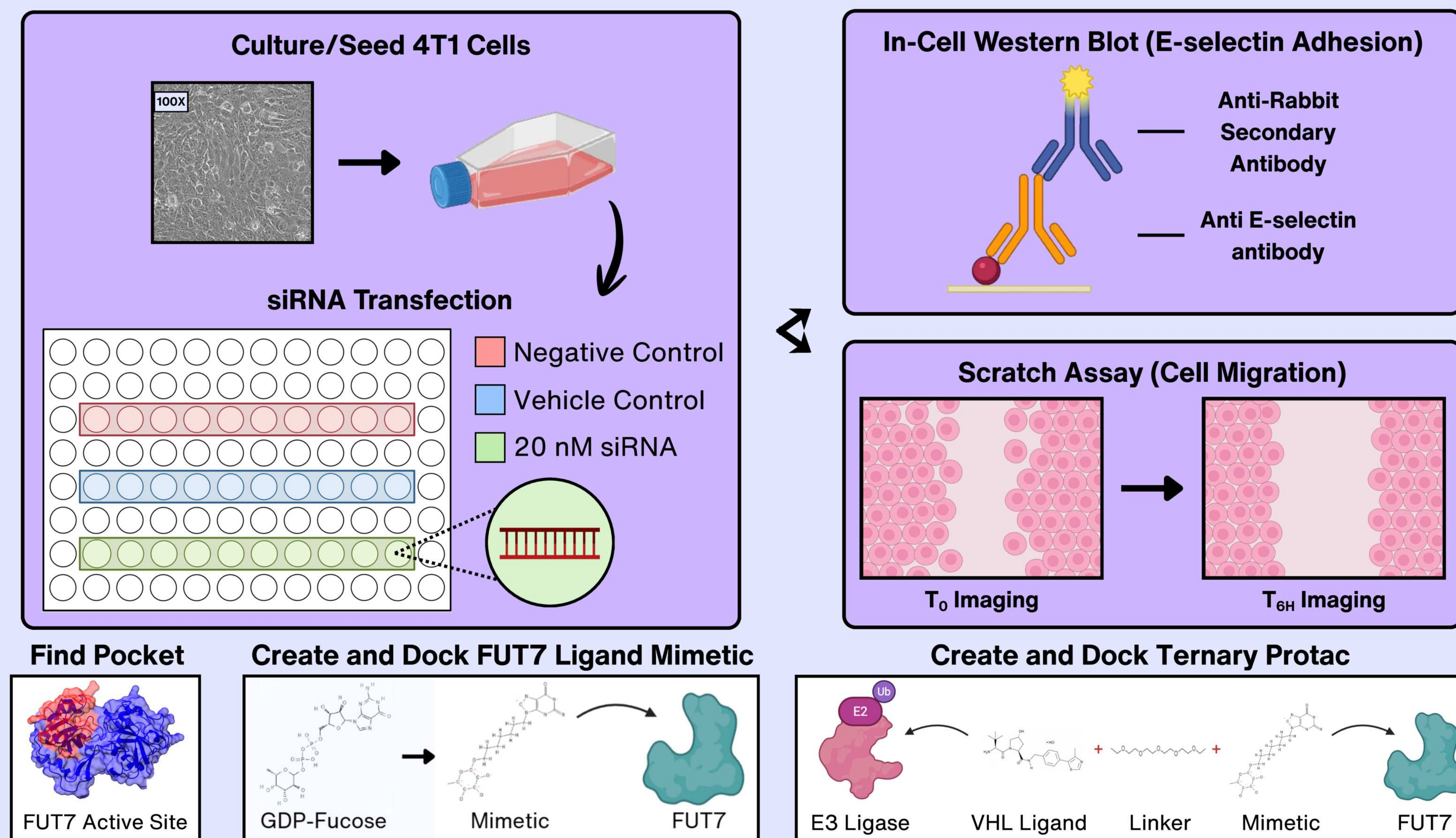


Figure 5. A visualization of the methodology used in the study. Graphic created by researcher using BioRender, 2026

Results

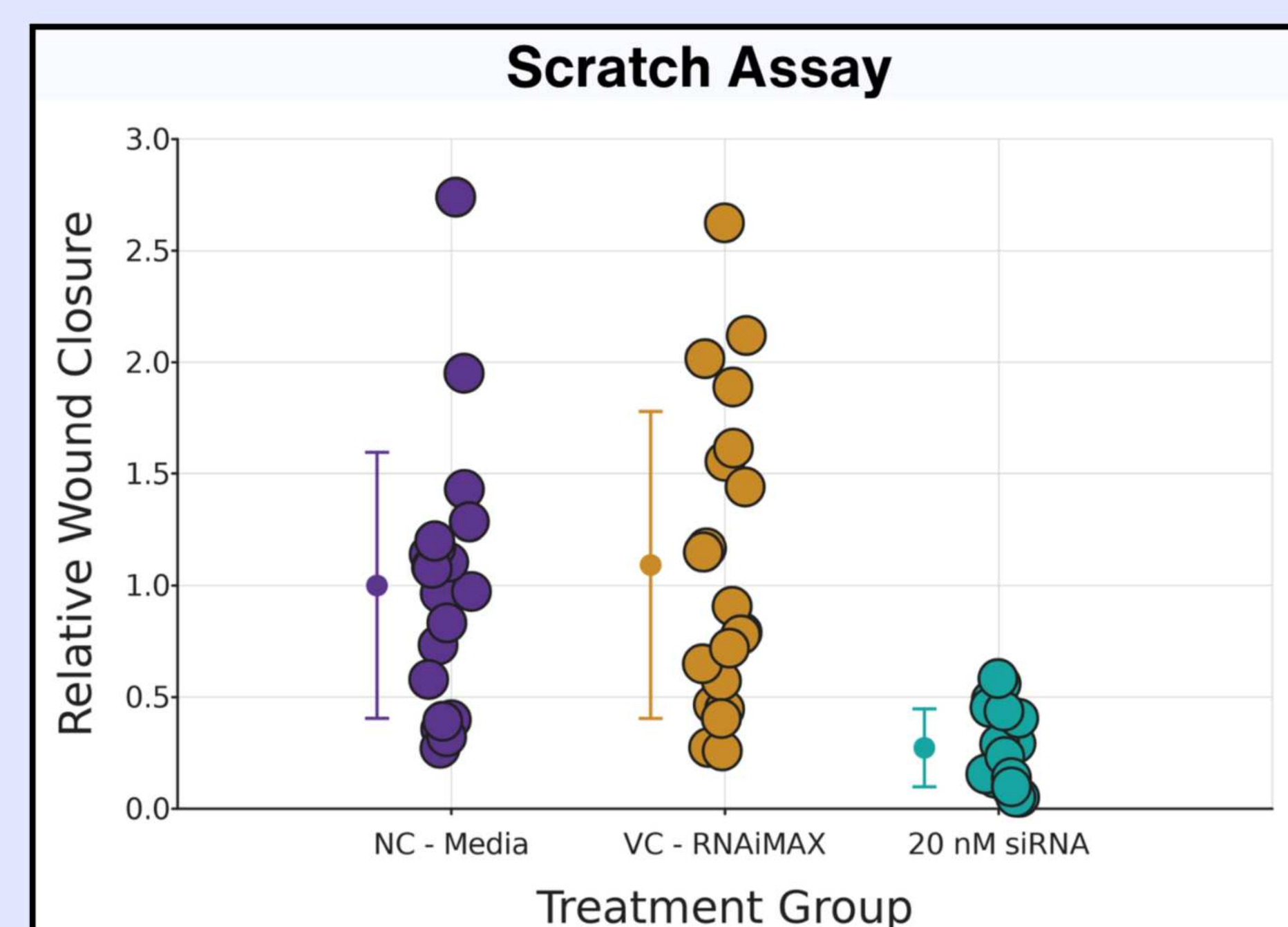


Figure 6. Migration of 4T1 cells after 6 hours in 4T1 cells. Treated groups were normalized to the negative control. Data represent pooled results from Trials 1 and 2. Data Visualization created by researcher using DataClassroom, 2026

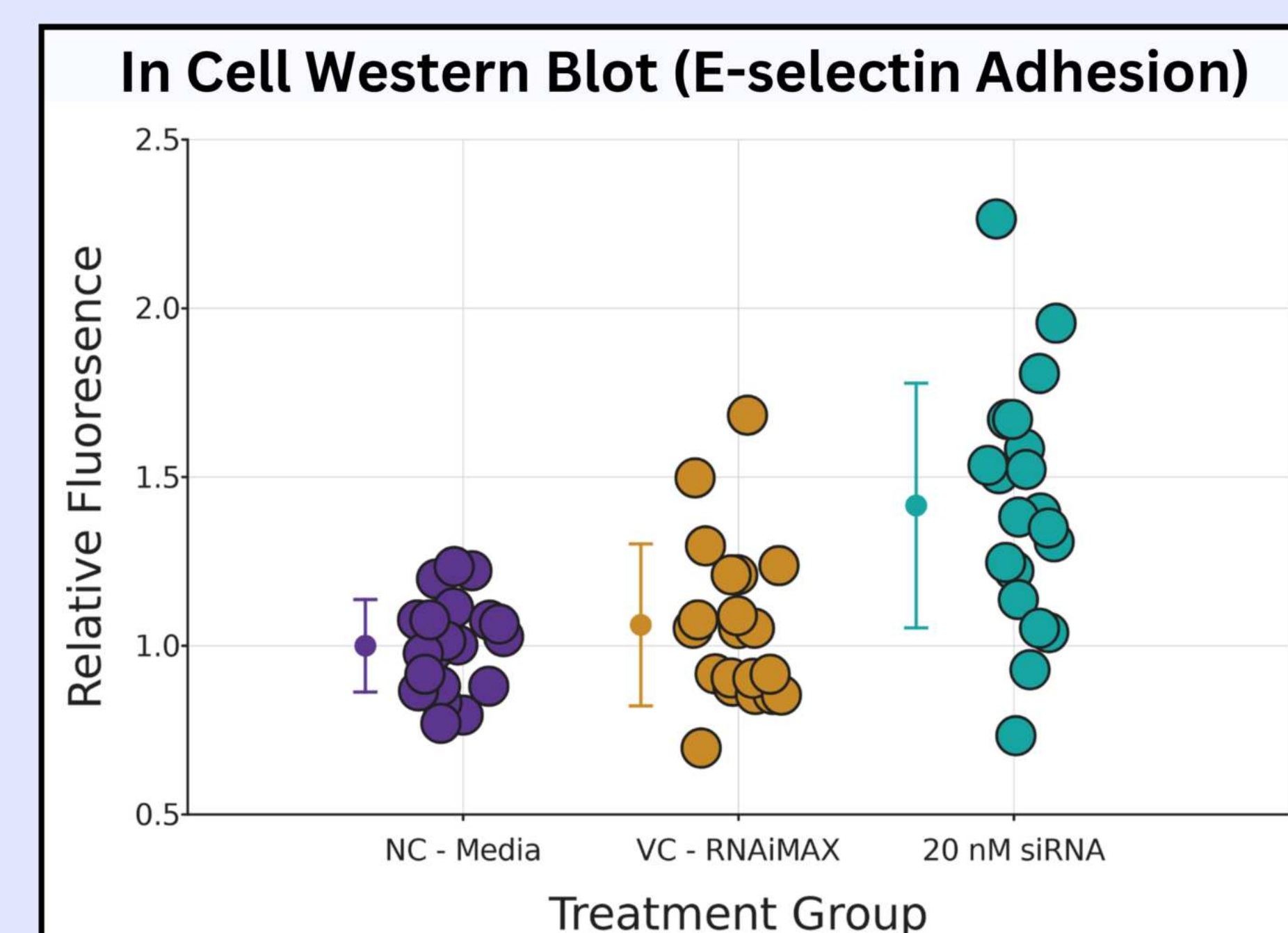


Figure 7. Recombinant E-selectin binding in 4T1 cells measured by in-cell western blot. Treated groups were normalized to the negative control. Data represent pooled results from Trials 1 and 2. Data Visualization created by researcher using DataClassroom, 2026

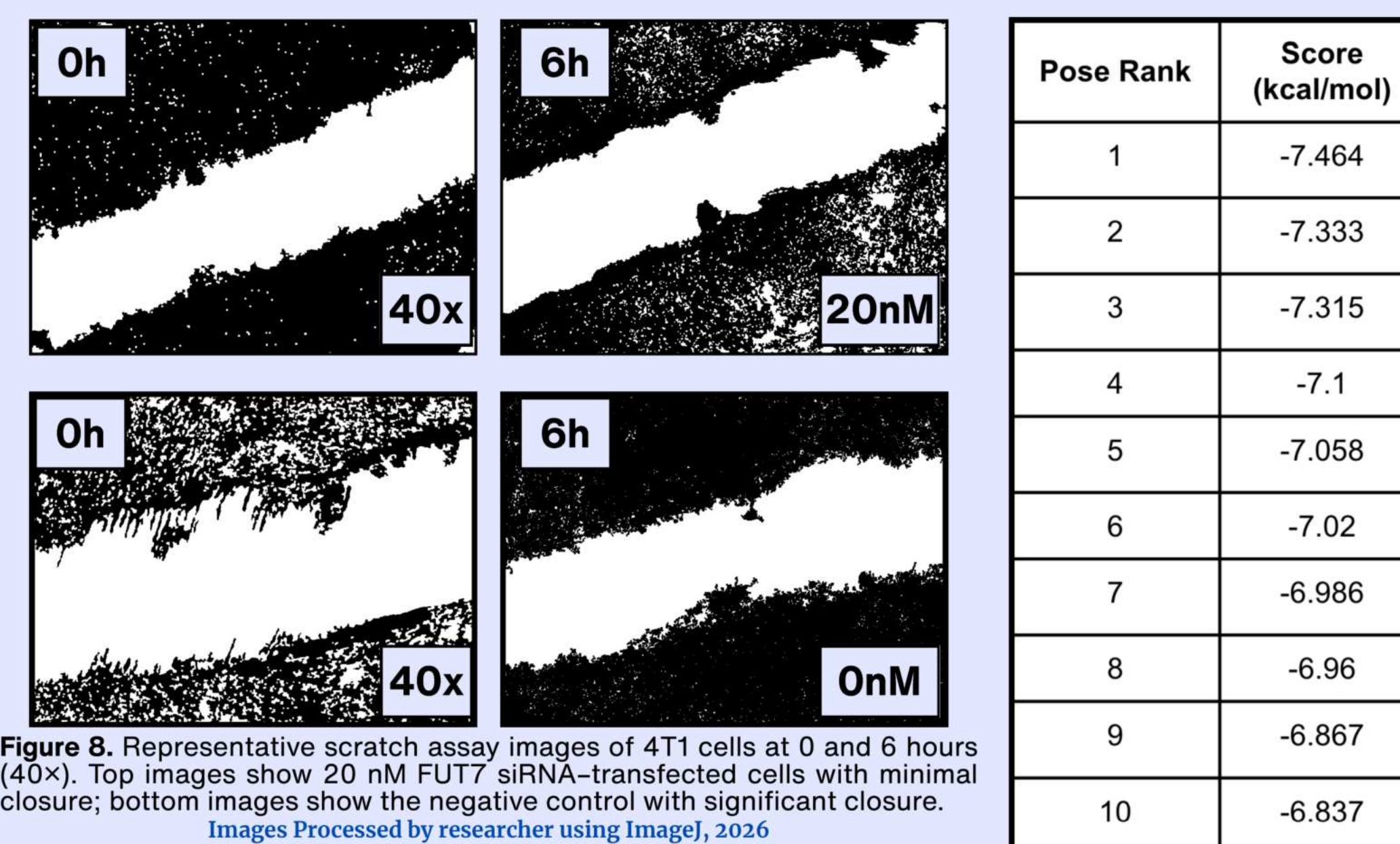


Figure 8. Representative scratch assay images of 4T1 cells at 0 and 6 hours (40x). Top images show 20 nM FUT7 siRNA-transfected cells with minimal closure; bottom images show the negative control with significant closure. Images Processed by researcher using ImageJ, 2026

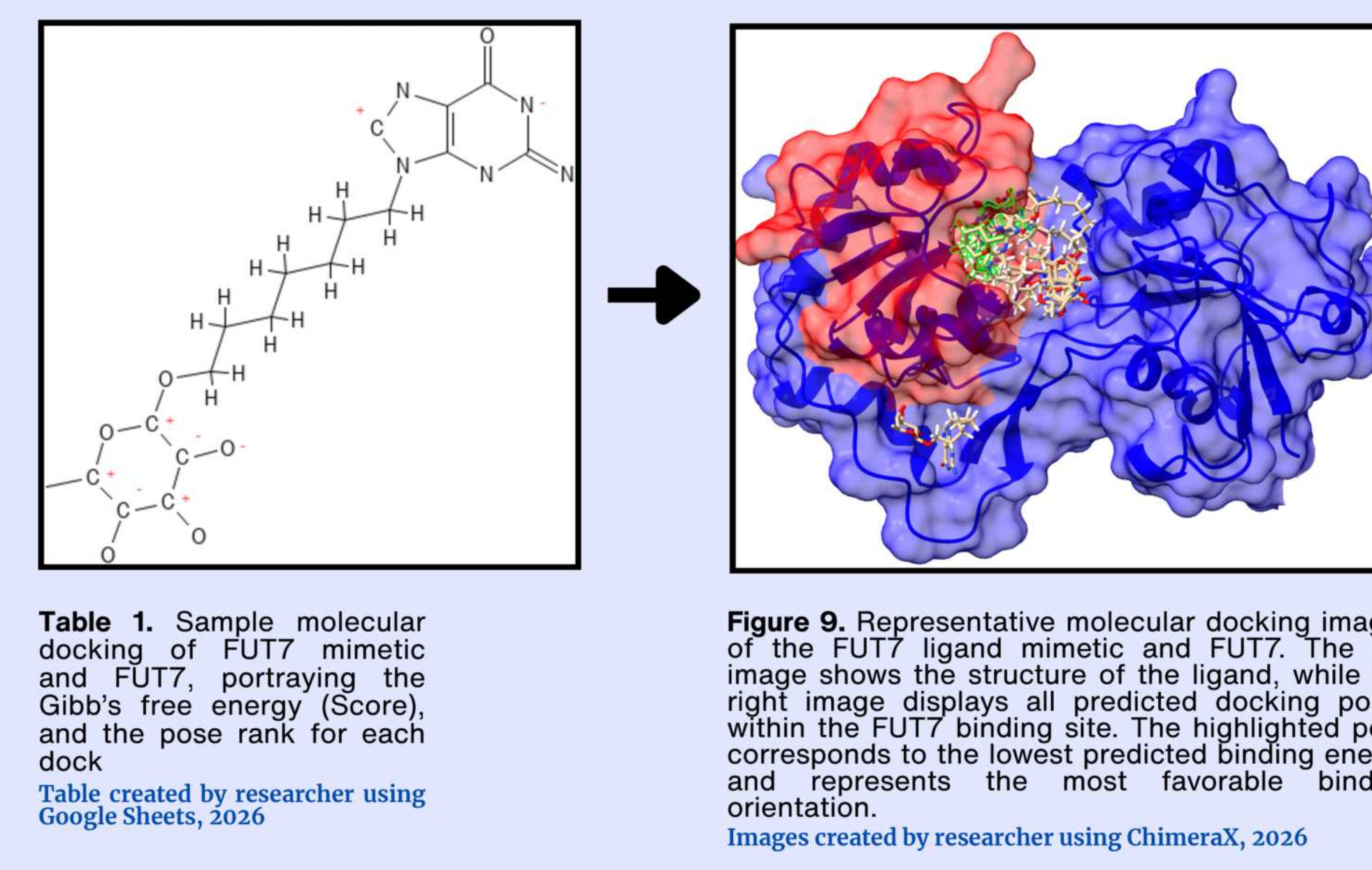


Figure 9. Representative molecular docking images of the FUT7 ligand mimetic and FUT7. The left image shows the structure of the ligand, while the right image displays all predicted docking poses within the FUT7 binding site. The highlighted pose corresponds to the lowest predicted binding energy and represents the most favorable binding orientation. Images created by researcher using ChimeraX, 2026

Discussion

Cell Migration

- **Data supported hypothesis:** FUT7 siRNA decreased 4T1 migration, suggesting FUT7-mediated glycosylation contributes to local invasion

Pair	P-Value	ANOVA P-Value
RNAiMAX - Media	0.85	<0.01
siRNA - Media	<0.01	
siRNA - RNAiMAX	<0.01	

Table 2. Significant ANOVA and post-hoc test values of normalized wound closure of 4T1 cells after 6 hours in 4T1 cells. Table created by researcher using Google Sheets, 2026

E-selectin Adhesion

- **Contrary to hypothesis:** FUT7 siRNA increased E-selectin binding; higher fluorescence indicates greater E-selectin adhesion in FUT7 knocked-down cells
- May reflect compensatory glycosylation pathways

Pair	P-Value	ANOVA P-Value
RNAiMAX - Media	0.74	<0.01
siRNA - Media	<0.01	
siRNA - RNAiMAX	<0.01	

Table 3. Significant ANOVA and post-hoc test values of normalized E-selectin binding measured by in-cell western blot. Table created by researcher using Google Sheets, 2026

FUT7 Ligand Mimetic Docking

- **Ligand mimetic successfully docked** within FUT7 active site
- **Strong binding affinity** suggests competitive inhibition

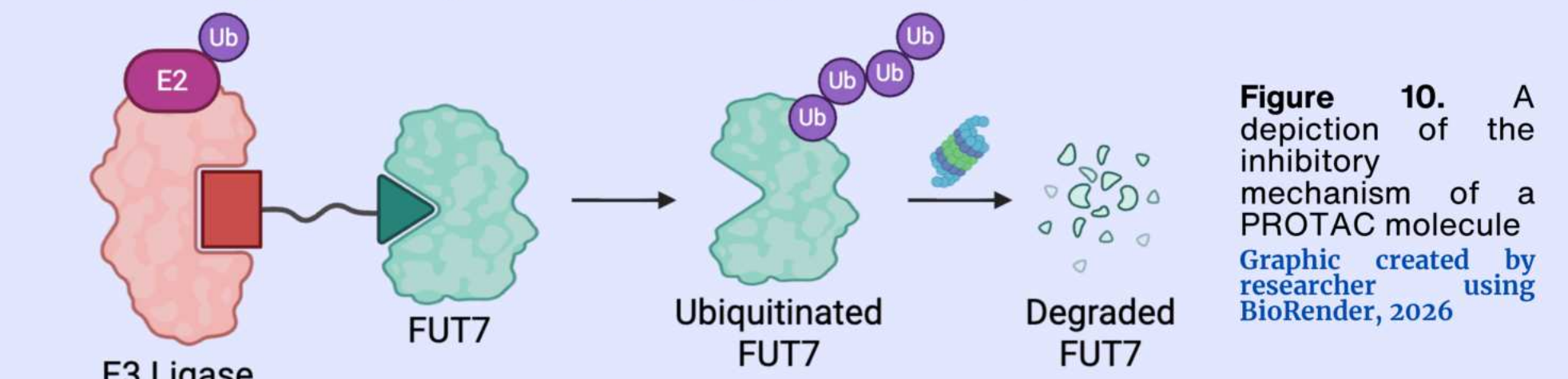


Figure 10. A depiction of the inhibitory mechanism of a PROTAC molecule. Graphic created by researcher using BioRender, 2026

Implications

- FUT7 may regulate glycosylation mechanisms involved in tumor invasion and adhesion
- Targeting FUT7 could disrupt glycan-mediated metastasis
- Potential therapeutic applications (PROTAC based strategy)

Future Work

The current findings provide a robust foundation for further investigation. Multiple avenues remain for extending this work:

- Extending the **in-cell western** analysis to include **L- and S-selectin** to further evaluate selectin-mediated interactions
- Evaluating down stream **effects on oncogenic signaling pathways** (MAPK, AKT, PI3K) and EMT-related transcription
- Developing **small carbohydrate-based inhibitors** of FUT7 by fine-tuning PROTAC designs in silico and validating their efficacy through synthesis and *in vitro* testing

Key References

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